YEAR 9 SCIENCE
CHEMICAL SCIENCES
TEST 1 - ATOMIC STRUCTURE, RADIATION

HOLY CROSS

NAME: SOLUTIONS **CLASS:** ______ **Mark:** —/26

All matter is made of atoms which are composed of protons, neutrons and electrons; natural radioactivity arises from the decay of nuclei in atoms.

Multiple Choice **Write the answer to each question in the appropriate box at right.**

1. γ, β, α, B

   IN ORDER, the above symbols mean:

   A Gamma, Beta, Alpha, Boron.
   B Gamma, Boron, Alpha, Beta.
   C Alpha, Beta, Gamma, Beryllium.
   D Alpha, Beryllium, Gamma, Beta.

2. The nucleus of which atom is identical to an ***alpha particle***?

   A H - Hydrogen
   B He - Helium
   C Li - Lithium
   D Ti - Titanium

3. Which of the following statements about atoms is ***true***?

   A An atom is mostly empty space.
   B Neutrons and protons are found in the nucleus with the electrons.
   C Electrons circle the nucleus all the time.
   D The number of neutrons found in the nucleus of an atom is always the same.

4. The ***mass number*** of an element is the total number of:

   A electrons in an atom.
   B protons in an atom.
   C protons and electrons in an atom.
   D protons and neutrons in an atom.

5. The ***atomic number*** of an element is the total number of:

   A electrons in an atom.
   B protons in an atom.
   C protons and electrons in an atom.
   D protons and neutrons in an atom.

QUESTION	ANSWER
1	A
2	B
3	A
4	D
5	B
6	C
7	A
8	B
9	D
10	A
11	A
12	A
13	D
14	A
15	B
16	B

6. An atom contains 4 protons, 5 neutrons and 4 electrons. Its mass number is:

A 4
B 5
C 9
D 13

7. An atom contains 4 protons, 5 neutrons and 4 electrons. Its atomic number is:

A 4
B 5
C 9
D 13

8. A charged atom is called:

A an isotope.
B an ion.
C a proton.
D radioactive.

9. Define ionising radiation:

A Radiation from the ionosphere.
B Radiation produced when atoms are ionised.
C Any type of electromagnetic radiation.
D Radiation that can remove electrons from atoms and molecules.

10. Identify the set that contains an element, a compound and a mixture.

A Carbon dioxide, air, silver.
B Gold, sulfur, phosphorus.
C Granite, oxygen, hydrogen.
D Copper oxide, calcium carbonate, sodium chloride.

11. 6. Beta particles are:

A fast-moving electrons.
B positively charged.
C heavier than an alpha particle.
D unable to penetrate human skin.

12. One of the particles inside the atom is very much lighter than the others. It is the:

A electron.
B neutron.
C nucleus.
D proton.

13. The nucleus of the atom contains:

A only neutrons.
B only protons.
C electrons and neutrons.
D neutrons and protons.

14. Which of the following are atoms of the same element?

	Number of protons	Number of neutrons	Number of electrons
Atom U	11	13	11
Atom V	13	11	13
Atom X	12	11	13
Atom Y	11	12	11

A U and Y
B U and X
C X and Y
D U and V

15. Different isotopes of the same element contain different numbers of:

A electrons.
B neutrons.
C protons.
D nuclei.

16. Iodine -131 has a half-life of 8 days. If a sample of I-131 has a mass of 60 g, how much radioactive iodine will be left after 16 days?

(a) 7.5 g
(b) 15 g
(c) 30 g
(d) 3.75 g

Short Answer **Write the answer to the questions in the spaces provided.**

1. Complete the following table.

ELEMENT	PROTONS	NEUTRONS	ELECTRONS
$^{19}_{9}F$	9	10	9
$^{56}_{26}Fe$	26	30	26
$^{39}_{19}K$	19	20	19

[$\frac{1}{2}$ mark off for each mistake]

(3)

2. What makes the nuclei of some atoms naturally unstable?

- There isn't enough force to hold the protons and neutrons together.
- There is too much energy in the nucleus.

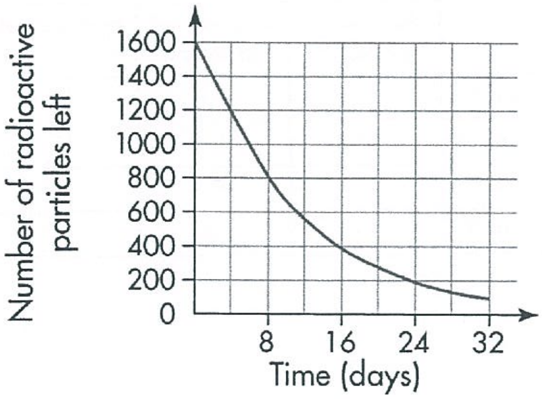
EITHER OK.

(1)

[Be lenient! Pay anything reasonable.]

(1)

3. The graph below shows the radioactive decay of iodine-131.



(a) What is the half-life of iodine-131?

8 days

(1)

(b) How many radioactive particles would be left after 40 days?

50

(1)

4. List ***two*** major long-term effects on humans caused by cell damage resulting from exposure to large amounts of nuclear radiation.

- Cancers - various types. • Permanent cell damage - burns, etc.
- Damage to DNA

[Any 2 reasonable effects]

(2)

5. Describe two uses of nuclear radiation in medicine and/or industry.

- Nuclear power
- Medicine - tracers, radiation treatment
- Industry - testing welds, tracing fluid movement
- Smoke detectors.

[Anything reasonable - 1 mark each]

(2)